Abhinav Java

Curriculum Vitae

Department of CSE
Delhi Technological University, Delhi
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Github *LinkedIn*

Education

2018 – Present **Delhi Technological University**, *New Delhi*, India,
BTech in Computer Science & Engineering | CGPA: 8.40/10.
Featured Coursework: Data Structures, Algorithm Design and Analysis, DBMS, Operating Systems, Computer Vision, Machine Learning and Deep Learning

2016 – 2018 **Amity International School**, *New Delhi*, India,
Senior Secondary Schooling, CBSE Board | Percentage: 93.2%.
Subjects: Physics, Chemistry, Mathematics, English, Physical Education

Research Experience

Sept. '20 – Present **Indian Institute of Technology**, *Bombay*,
Deep Learning Research Intern.

- Conducted a literature study of 3D computer vision with a focus on reconstruction applications.
- Developed a text image restoration pipeline using semantic encoding loss and a generative strategy allowing robust cross domain performance in the field of optical character recognition with only 500 target data examples. **(Manuscript in preparation)**

Supervisor: **Dr. Ganesh Ramakrishnan**, *Professor, Department of Computer Science & Engineering*, IIT Bombay (*Personal Web-page*)

Apr. '20 – Present **Delhi Technological University**, *New Delhi*,
Undergraduate Research Assistant, Machine Learning Lab.

- Designed and implemented a transformer based image dehazing pipeline which has able to achieve state of the art performance on multiple benchmarks.
- Did a thorough literature survey of most representative deep learning based solutions to understand and evaluate the current state of image dehazing algorithms.

Supervisor: **Dr. Anil Singh Parihar**, *Associate Professor, Department of Computer Science & Engineering*, DTU (*Google Scholar*)

Industry Experience

May – Jul. '20 **IDfy**, *Mumbai*,
Computer Vision & Machine Learning Intern.

- Developed an offline signature verification microservice for document validation. Trained squeezenet based Siamese network on publicly available datasets and initiated its beta release towards the end of internship period.
- Initiated the integration of static and security checks in CI/CD pipelines using bandit and pylint. Did extensive testing using JMeter and Insomnia to ensure robustness of existing python apps.

Supervisor: **Mr. Gurudatt Bhobe**, *Manager, ML@IDfy*

Publications

Under Review

Jan. '21 **Number Theory in Neural Networks**.
Made a striking connection between generalization of trained models and the correlation between their weights and behavior of numerical data. Carried out several experiments to show generalization and proposed an early stopping criteria that doesn't need a validation set. **(Anonymous for double blind review)**

2020 **Abhinav Java***, **Kavinder Singh***, **and Anil Singh Parihar***, Comparative Analysis of Deep Learning based Image Dehazing.

Communicated Journal Article

2020 **Anil Singh Parihar***, **Kavinder Singh***, **and Abhinav Java***, DehazeFormer: Image Dehazing using Transformers.

Projects

Nov. '20 **Image Captioning using Multimodal Semantic Similarity**, *GitHub*.
Implemented a loss function which maps the word embedding and image features to a common space and optimizes the similarity between the two vectors as part of ML project. The proposed loss outperforms the baseline model on perplexity score and converges faster.

Aug. '20 **Handwritten Formreader**, *GitHub*.
Developed an end to end attention based OCR pipeline to detect text in boxed forms. The model was trained only on synthetic data generated using open source tools like text-renderer and still managed to achieve high accuracy on real data. The REST API was built using flask and is deployed on heroku.

Nov. '19 – Jan. '20 **Multi Agent Multi Target Swarm Search**, *UAS*, DTU.
Implemented a custom multi object search for 25 drones both in a centralized and decentralized fashion for the efficient human search using UAVs. This was simulated on the DroneKit Python API. The algorithm was tested on college ground and with the Indian Airforce at their base in Pokhran.

Professional Responsibilities and Leadership

Sept. '19 – Oct. '20 **Unmanned Aerial Systems**, ***DTU***, Indian Air Force Mehar Baba Prize.
- Worked with a team of around 20 highly motivated undergraduates towards development of a swarm of 50 UAVs with capabilities of extreme weather tolerance, 50km range, human detection, payload drop etc for aid in Humanitarian Assistance and Disaster Relief situations for the Air Force.
- Practiced rigorous testing of multiple systems at Indian Air Force Base to validate these algorithms.

Jan. '16 – Mar. '17 **High School Council**, *Additional Cultural Secretary*, Amity International School, Noida.
- Managed several high school proceedings and played the Tabla in all major school events.

Achievements & Recognitions

2018 – 2020 Partnered with **Adani Defense and Aerospace**, UAS-DTU is one of the top 5 and currently the only undergraduate Research and Development team in the development process of IAF-Mehar Baba Prize, Phase 3. The prize for engineering a robust aerial swarm in the previous phase was INR 25 lakh.

2019 Stood first at Aavahan (Sports Fest DTU), Udgosh (Sports Fest IIT Kanpur) and Triquetra (Sports Fest IIITD, DTU, NSIT) during the 2018-2019 football season.

2018 Achieved 96.54^{th} percentile (5500 All India Rank) in **Joint Entrance Examination (JEE) – Advanced** out from about a hundred thousand students getting selected for this exam annually.

2018 Achieved 99.35^{th} percentile (7753 All India Rank) in the **Joint Entrance Examination (JEE) – Main**, out from about 1.3 million students all over India.

2016 One of the only two students in the batch to qualify for the **National Talent Search Examination (NTSE)** stage 2 in high school.

2015 Participated in **JENESYS Youth Football Exchange** program to Japan as a sports-cultural representative from India.

Skills

Languages & Frameworks: Python (proficient), C (familiar), C++ (familiar), PyTorch, Keras, Tensorflow, Pandas, Numpy, OpenCV, Bash Scripting, Octave, Flask, Django, SQLite

Tools: Colab, GCP, Insomnia, JMeter, GIT, LaTeX

Academic References

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* Denotes Equal Contribution
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